

ROK

रोक · நிறுத்து — to stop

Stopping the money before the paperwork starts.

A rethink of India's National Cyber Crime Reporting Portal, designed so that the fastest path through it is also the accessible one.

Government platform	National Cyber Crime Reporting Portal — cybercrime.gov.in
Owning body	Indian Cyber Crime Coordination Centre (I4C), Ministry of Home Affairs
Core problem	The Golden Hour is spent filling the form
Deliverable	Working browser-based proof of concept
Prepared by	Jananth Nikash K Y
Date	24 August 2026
Version	1.0 — Concept & POC Design Document

Executive Summary

When money is stolen online in India, the machinery that gets it back already works. The Citizen Financial Cyber Fraud Reporting and Management System, launched in 2021, connects police, banks, wallets and telecom operators so that a single complaint can trigger a hold on the beneficiary account while the funds are still traceable. The Government has told Parliament that **more than ₹7,000 crore has been saved across more than 23 lakh complaints** through this system.

The bottleneck is not the police, the banks or CFCFRMS. It is citizen intake. To open a financial-fraud complaint on cybercrime.gov.in, a victim must register with a mobile number and an OTP, classify their own crime into a government taxonomy, select the State or Union Territory that will investigate, locate a twelve-digit transaction ID or UTR, scan and upload a government photo ID, and write an incident description of at least two hundred characters containing no special characters. The portal is available in Hindi and English only.

That is the sequence the state asks of a person in the worst hour of their year. It is a sequence that a sixty-eight-year-old Tamil speaker, a blind user, a person who cannot read fluently, or anyone still shaking from the phone call cannot complete in time. And an incomplete form on NCRP is not a partial complaint — **it is no complaint at all**. Every minute lost here is money that leaves the mule account.

Rok re-architects what a fraud report is. It opens a timestamped case on the very first tap, before anything is known. It never asks for a UTR — instead it shows the victim their own bank messages, reads them aloud in their language, and asks one question: *“which one is the wrong one?”* It infers the government taxonomy from four icon taps. It composes the mandatory two-hundred-character description itself and reads it back for a yes or no. A visible Golden Hour clock governs the interface, permitting on screen only the questions that change whether the money gets frozen in the next sixty minutes; everything that is paperwork is deferred until after the case exists.

The result is a complete, submittable financial-fraud complaint in under a minute, with **zero words typed and no English required**. Critically, the accessible path and the fast path are the same path — which is the strongest argument for universal design that a demonstration can make.

At a glance	
The inversion	NCRP is a prove-it document, structured for the investigator who reads it later. Rok is a stop-it signal, structured for the money that is moving now.
Unit of success	NCRP measures completed complaints. Rok measures an opened case plus one freeze-relevant identifier — the only part the Golden Hour cares about.
Accessibility strategy	Recognition of your own evidence, not translation or added voice. Recognition survives panic, age, low literacy, blindness and unfamiliar language.
Time to complaint	Under 60 seconds, versus a four-part form documented at roughly 21 steps.
Failure mode	Give up halfway and a timestamped case still exists, ready for a helper or a 1930 operator to complete.
POC honesty	NCRP has no public API. Rok produces a validated submission packet and a 1930 read-aloud card — not a faked integration.

Contents

Executive Summary.....	2
1. Chosen Government Platform.....	4
2. The Real Problem.....	4
3. Why the Existing Experience Fails.....	6
4. Target Users.....	7
5. One Unique Solution: Rok.....	7
<i>Figure 1 — System architecture.....</i>	<i>10</i>
6. Step-by-Step User Journey.....	11
7. What Makes It Different.....	11
<i>Figure 2 — The 60-second journey.....</i>	<i>12</i>
<i>Figure 3 — The inversion.....</i>	<i>13</i>
8. Why This Could Stand Out in the Hackathon.....	14
9. Browser POC Scope.....	14
10. Key Screens and Interactions to Build.....	15
11. Risks and Limitations.....	16
12. One-Sentence Pitch.....	16
Appendix A — Evidence Register.....	17
Appendix B — Field Mapping.....	17
Appendix C — Build Plan.....	18

1. Chosen Government Platform

National Cyber Crime Reporting Portal — cybercrime.gov.in

Operated by the Indian Cyber Crime Coordination Centre (I4C), Ministry of Home Affairs. Companion helpline: 1930.

1.1 Why this platform, and not the other nine

All ten platforms were examined against four filters: how frequently ordinary Indians touch them, how badly the experience fails for a user with low literacy or a disability, whether the failure causes irreversible harm, and whether a browser POC can demonstrate a real improvement. Only one platform scores at the top of all four.

THE DECIDING FILTER

NCRP is the only platform on the list where a user's age, language, literacy or disability converts directly into money lost within the next sixty minutes. Everywhere else, a bad experience costs a bad afternoon.

Platform	Assessment	Verdict
IRCTC	High volume and genuine frustration, but users are habituated and failure is recoverable — you book the next train.	Set aside
Income Tax e-Filing	Annual, deadline-driven, and most low-literacy users already delegate to an intermediary.	Set aside
GST	Business-facing, mediated almost entirely by accountants. The accessibility brief barely applies.	Set aside
MCA	Professional filing platform. Near-zero overlap with the target user groups.	Set aside
Parivahan Sewa	Real pain in licence and RC journeys, but slow-clock and heavily mediated by agents.	Set aside
EPFO	Painful, high-stakes claim rejections for exactly the right users — but the harm unfolds over weeks, so a demo cannot show urgency.	Strong, set aside
RTI Online	Excellent civic value, but a niche of motivated users and an inherently slow clock.	Set aside
UMANG	An aggregator. Fixing it means fixing the services underneath it — too diffuse for one POC.	Set aside
CPGRAMS	Runner-up. Genuine misrouting and boilerplate-closure problems, but the harm is diffuse and the demo is undramatic.	Runner-up
National Cyber Crime Reporting Portal	Frequent, catastrophic, irreversible, disproportionately harms the exact user groups in the brief, and the clock is measured in minutes.	SELECTED

2. The Real Problem

THE PROBLEM, IN ONE LINE

The Golden Hour is spent filling the form.

2.1 What the state already does well

India's fraud-freeze machinery is genuinely capable. The Citizen Financial Cyber Fraud Reporting and Management System (CFCFRMS), operational since 2021 under I4C, exists specifically to enable immediate reporting of financial fraud and to stop the siphoning off of funds. It is supported by the Samanvay data-sharing platform and the Cyber Fraud Mitigation Centre, which put police, banks, payment aggregators and telecom operators in the same room. In a written reply to the Rajya Sabha in December 2025, the Government stated that more than ₹7,000 crore has been saved across more than 23 lakh complaints through this system.

The lever works. What fails is the handle a citizen has to pull.

2.2 What the portal demands, in the first minutes

The following is not a critique of the interface. It is the portal's own published list of what a victim must supply.

What is required	Why it defeats the user in a crisis
Registration with name and Indian mobile number, then an OTP valid for 30 minutes	A gate before the complaint even begins — and a fraud victim has just been trained, minutes earlier, to fear anything asking for an OTP.
Self-selection of complaint category and sub-category	The citizen is made responsible for legal classification while panicking.
Selection of the State / UT that will handle the complaint	The citizen is made responsible for jurisdictional routing.
A 12-digit transaction ID / UTR number	A pure recall-and-typing task — while the answer sits verbatim in an SMS on the same phone.
Name of the bank / wallet / merchant, transaction date, fraud amount	Four more fields, all of which are already inside that same SMS.
A soft copy of a national ID (Voter ID, DL, Passport, PAN, Aadhaar), under 5 MB	Requires a scan or a good photo — minutes that the money does not have.
An incident description of minimum 200 characters, with no special characters	The victim must write an essay about their own humiliation, and must avoid apostrophes while doing it.
A four-part form (Incident, Suspect, Complaint, Preview) — documented as roughly 21 steps	Twenty-one opportunities to stop.
Hindi or English	In a country with twenty-two scheduled languages, for a crime that disproportionately hits people who read neither.

2.3 And the portal knows

The NCRP financial-fraud form contains a field labelled “Is there any delay in reporting?”. The system understands perfectly well that delay is what kills recovery. Its only response is to ask the victim about it afterwards.

2.4 The scale

Figure	Detail	Source
₹7,769.63 cr	Financial cyber fraud complaints lodged in 2025 by 1,03,488 senior citizens and 4,63,114 women combined.	MHA reply, Rajya Sabha (Aug 2026)
₹4,005.12 cr	Of that total, the share lodged by senior citizens alone — 1,03,488 complaints.	MHA reply, Rajya Sabha (Aug 2026)

Figure	Detail	Source
₹22,845 cr	Total cyber-fraud losses in 2024, up from ₹7,465 cr in 2023; cases rose from 24 lakh to 36 lakh.	MHA data placed in Parliament
₹7,000 cr+	Saved across 23 lakh+ complaints through CFCFRMS since 2021.	MoS Home, Rajya Sabha (Dec 2025)
12.21%	Share of reported cybercrime cases in Goa with elderly victims in 2025 — up from 6.17% in 2023.	Goa Police data, reported Jan 2026

3. Why the Existing Experience Fails

Not as a list of interface complaints — as five structural failures, plus one that is not about interfaces at all.

3.1 It is all-or-nothing

An incomplete NCRP form is not a partial complaint. It is no complaint. The one moment at which a user is least capable of completing anything is the one moment at which completion is mandatory. This is the single largest design flaw, and it is invisible because nobody measures abandoned forms.

3.2 It asks for recall, not recognition

“Enter your 12-digit UTR” is a memory task, a search task and a typing task stacked on top of each other, in a script the user may not read. The victim is holding a phone that already contains the answer, verbatim, in a bank SMS.

3.3 It makes the victim the classifier

Category, sub-category and State/UT are all selected by the complainant, and complaints are then handled by the State/UT police based on that selection. A citizen in shock is made responsible for the correct legal and jurisdictional routing of their own case.

3.4 Its accessibility is a link, not a design

The portal’s accessibility control offers a high-contrast and a normal-contrast toggle. Its provision for screen-reader users, listed under the Learning Corner, is a link to download NVDA. Its language switch offers Hindi and English. This is compliance as a checkbox, and it collapses entirely under the conditions in which the portal is actually used.

3.5 The parallel channel is congested too

The 1930 helpline is the fast lane, and it is crowded. Karnataka State Police publicly upgraded the helpline with an IVR and a voice-guided WebBOT precisely because callers were abandoning the queue: if a complainant cannot reach a staff member or disconnects due to long wait times, an SMS with a link to the WebBOT is sent automatically, and helpline staff later enter the information into NCRP manually. That is a state police force conceding that the front door does not open fast enough — and then bolting a second, slower door onto the same form.

3.6 Underneath all of it: fear, shame and isolation

Digital-arrest frauds work by keeping the victim on a continuous call, applying psychological coercion, and explicitly instructing them to tell nobody. Victims have been placed under secrecy oaths and kept in silence for days. In March 2025, an elderly couple in Karnataka died by suicide after being defrauded of ₹50 lakh. A form is not a response to that. Neither is a chatbot.

THE DESIGN CONSEQUENCE

Any solution that only makes the existing form prettier, translated, or read aloud is treating a triage problem

as a presentation problem. The questions themselves have to go.

4. Target Users

Rok is designed for six people at once. The test of the design is that none of them needs a different version of the product.

User	What breaks today	What Rok changes
Senior citizen, low digital confidence	Cannot complete an OTP-gated, four-part, 21-step form fast enough; often does not know 1930 exists.	One tap opens the case. Four questions, spoken aloud, in large type.
Blind or low-vision user	A contrast toggle and a link to download NVDA. Free-text fields and dropdown taxonomies are hostile to screen readers.	Every state emits audio. The core question is answered by ear: “message two, forty-five thousand rupees at 11:47 PM — that one.”
Deaf or hard-of-hearing user	Excluded from the 1930 helpline entirely; must use the slower written channel.	The full flow is visual and textual, with complete parity. No step requires hearing anything.
Person who cannot read or write fluently	A 200-character written description is an absolute barrier.	Nothing is written. Icons and recognition only; the description is composed for them and read back.
Non-English, non-Hindi speaker	The portal offers Hindi and English.	The interface speaks their language; the submission is generated in police-readable English.
Rural user with low digital literacy	Category taxonomies, jurisdiction selection and ID scanning are unfamiliar in combination.	Taxonomy inferred; jurisdiction inferred; ID deferred to after the freeze request.
The helper — son, neighbour, bank clerk, CSC operator	The portal silently assumes the person typing is the person robbed. There is no concept of assisted reporting.	Guardian handoff: a short code lets a trusted person continue the same live case. The victim stays the owner.
Confident, literate, experienced user	Is also panicking, and is also slow.	Same flow. Twelve seconds instead of twelve minutes.

5. One Unique Solution: Rok

Rok means stop. Not “file a complaint”. Stop the money.

5.1 The core inversion

THE IDEA

The NCRP form is a prove-it document, structured for the investigator who will read it later. Rok is a stop-it signal, structured for the money that is moving right now. Everything time-critical is collected in the first forty seconds. Everything that is paperwork happens after the case already exists.

This single reframe generates every other decision in the product. It also changes the unit of success: NCRP counts completed complaints; Rok counts an opened case plus one freeze-relevant identifier. That is a smaller, faster, achievable target — and it is the only part of the report the Golden Hour actually cares about.

5.2 The six mechanics

Each mechanic is simultaneously the accessibility strategy. None of them is “add voice”.

#	Mechanic	How it works, and why it is also the accessibility answer
M1	The case opens on the first tap	One button, one spoken sentence, or one short code. A timestamped case exists before a single detail is known, and nothing the user does afterwards can destroy it. This flips the failure mode: today, giving up means nothing was reported; in Rok, giving up leaves a case a helper or a 1930 operator can complete.
M2	Point, don't type	Rok never asks for a UTR. It shows the user their own bank messages — large, one at a time, read aloud in their language — and asks a single question: “Which one is the wrong one?” It then extracts amount, timestamp, bank, account tail, UTR/RRN and beneficiary handle from the message itself. This converts the hardest task into recognition of something the user already knows is wrong. Recognition is the one cognitive function that survives panic, age, low literacy, blindness and unfamiliar language.
M3	The clock is the information architecture	A visible Golden Hour countdown does not decorate the screen — it decides what the screen is permitted to ask. Only questions that change whether the money is frozen in the next sixty minutes may appear. ID upload, address, police station, taxonomy and the 200-character essay are automatically deferred to a calm, post-freeze phase. Every question states what it buys: “this is the number that tells the bank which money to stop.”
M4	The narrative is derived, never written	The mandatory description is composed from the structured facts the user confirmed, in clean police-readable English, stripped of special characters. In return, Rok reads three plain-language sentences back in the user's own language, one at a time, for a yes or no on each. The same step is the audio path for blind users, the text path for deaf users and the confirmation path for everyone.
M5	“Are they still on the phone?”	The very first screen asks it. If yes, Rok does not ask for a UTR — it says, aloud: hang up now, do not tell them, send nothing more, and offers one tap to alert a trusted contact. This is a direct counter to documented digital-arrest isolation, and no existing government front door does it.
M6	Guardian handoff, and nothing worth phishing	One tap generates a short code or QR. A trusted person opens the same live case on their own phone and continues it, while the victim remains the case owner. Rok never requests an OTP, PIN, password or full account number — from anyone, ever — and says so in a permanently pinned line: “We will never ask for your OTP. Anyone who does is a fraud.”

5.3 Testing the idea against its own brief

Challenge	Answer
Is it a chatbot?	No. There is no free-text conversation, no open prompt and no “how can I help you”. It is a deterministic four-question spine with recognition-only inputs. The parsing is invisible plumbing, not the product.
Is it a translation layer?	No. Translating a hostile form yields a hostile form in Tamil. Rok removes the questions rather than translating them.
Is it “just add voice”?	No — and this matters, because Karnataka already added voice to the same form. The WebBOT exists because the form is too slow, and it did not make the form shorter.

Challenge	Answer
Does it already exist?	No prior art was found for evidence-first extraction into an NCRP-shaped complaint, in government or private tooling. That is a search result, not a proof of absence, and it is stated as such.
Would it genuinely improve outcomes?	Yes, if it shortens time-to-first-freeze-signal. That lever is real and is quantified by the Government's own CFCFRMS savings figures.
Does it respect government constraints?	Yes. It does not ask I4C to rebuild NCRP. It is an intake layer in front of machinery that already works, and it fabricates no integration it does not have.

5.4 System architecture

Rok is a four-layer system. Layers 1–3 are the proof of concept and run entirely in the browser; Layer 4 is existing Government of India infrastructure and is not modified. Figure 1 overleaf shows the full architecture.

Layer	Component	Responsibility
1	Citizen input	Six equivalent entry points — a tap, a spoken sentence, a bank SMS, an uploaded screenshot, an icon tap, or a guardian short code. Any one is sufficient to open a case.
2	Case Opener	Creates and timestamps the case on first interaction. Never blocked by missing data.
	Safety Triage	The “still on the phone” branch; hang-up script and trusted-contact alert.
	Golden Hour Orchestrator	Holds the clock and gates which questions the interface may render.
	Evidence Recognition Engine	Deterministic Indian bank-SMS parser plus OCR. Emits amount, timestamp, bank, account tail, UTR/RRN, beneficiary VPA, with a confidence flag per field.
	Taxonomy Inference	Maps four icon taps to NCRP category, sub-category and State/UT.
	Narrative Composer + Read-Back	Generates the ≥200-character, special-character-free description; drives the sentence-by-sentence confirmation.
	Accessibility Bus	Cross-cutting. Every state change emits audio, text and icon simultaneously; owns ARIA-live regions, focus order, TTS and STT.
	Local Case Store	Browser-local persistence. No OTP, PIN, password or full account number is ever requested, stored or transmitted.
3	Output adapters	1930 read-aloud card · NCRP submission packet · case PDF · guardian handoff code.
4	Government layer (unmodified)	1930 / I4C · cybercrime.gov.in · the victim's bank and cyber police station · CFCFRMS, Samanvay and the Cyber Fraud Mitigation Centre, which place the hold.

FIGURE 1

Rok — System Architecture

A client-side citizen-intake layer placed in front of the Government of India freeze machinery that already works

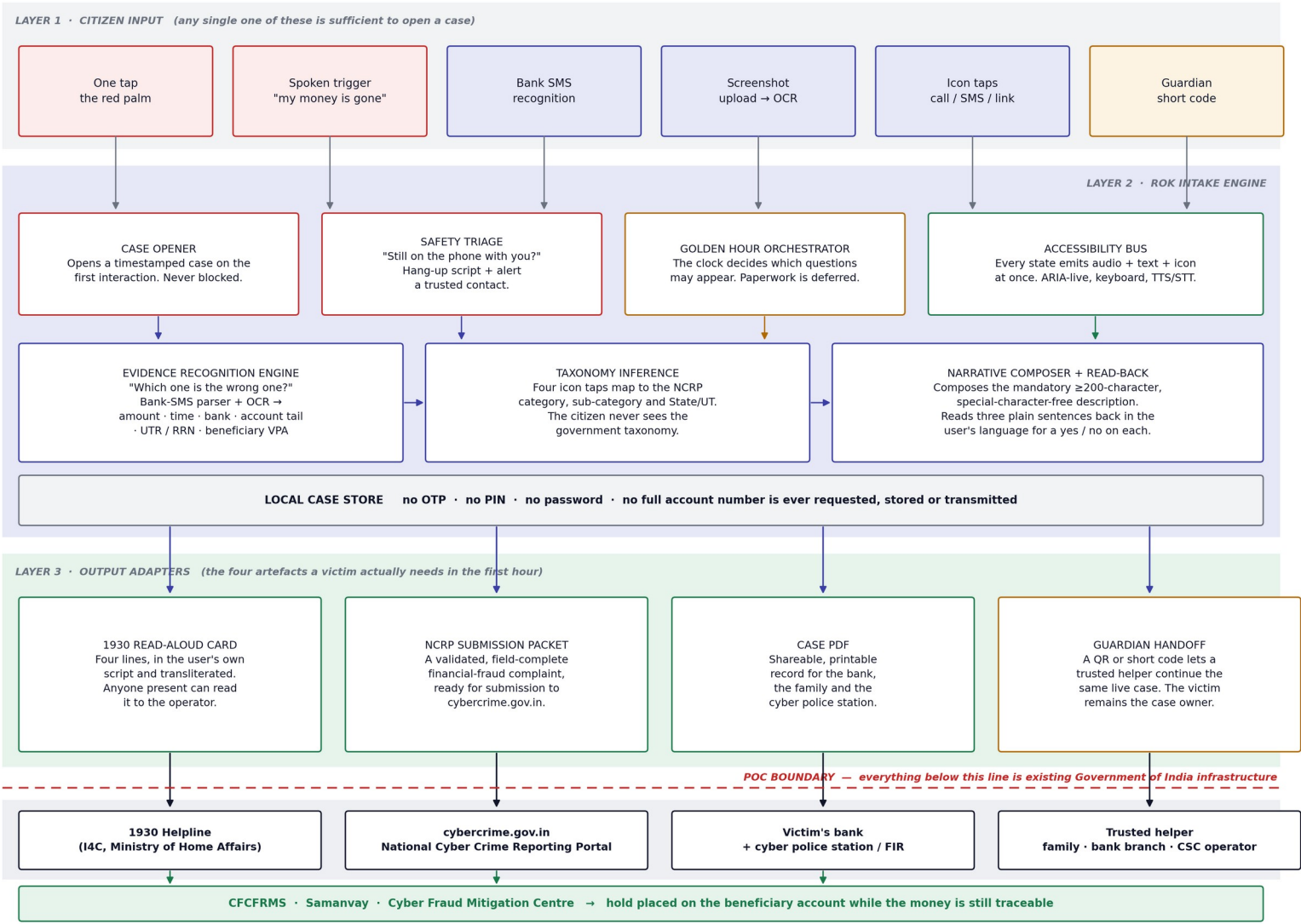


Figure 1 — Rok system architecture. Layers 1–3 are the browser-based proof of concept; everything below the dashed boundary is existing Government of India infrastructure.

6. Step-by-Step User Journey

Kamala, 68, Chennai. Tamil speaker, reads a little. ₹45,000 taken at 11:47 PM.

Time	What Kamala does	What Rok does
0:00	Opens Rok. One red palm-print button fills the screen, captioned in Tamil: “My money is gone.” She taps it.	The case opens. The clock starts. Nothing has been asked yet.
0:04	Hears, in Tamil: “Are they still on the phone with you?” Two large icons. She taps No.	If Yes: an immediate spoken hang-up instruction and a one-tap alert to a relative. Branch closed.
0:10	Sees her last five bank messages as large cards, each read aloud in Tamil. She taps the third.	Parses ₹45,000, 11:47 PM, bank name, account tail, UTR and beneficiary VPA. Zero typing.
0:22	Hears: “Only this one, or were there more?” Taps “Only this one”.	Builds the debited-transaction table.
0:30	Hears: “How did they reach you?” Four icons — call, SMS, WhatsApp, link. She taps the call icon.	Infers NCRP category and sub-category. She never sees the taxonomy. Infers State/UT from locale, then confirms it.
0:38	Hears a three-sentence Tamil summary of what happened. Taps yes after each one.	Composes the ≥200-character English narrative, special characters stripped. The complaint is now complete.
0:45	The screen turns green. One number, very large, read aloud: her case reference.	Generates the 1930 read-aloud card (four lines, Tamil script and transliteration), the validated NCRP packet, and a shareable PDF.
0:50	Sees three cards in order: “Your money”, “Your job now”, “By tomorrow”.	Hold requested · bank fraud line pre-dialled · nearest cyber police station mapped. The case card stays live.
Later	Prompted once, calmly: “When you’re ready — a photo of any ID card.”	Attaches the ID, accepts further evidence, completes the deferred paperwork.
THE RESULT <i>A complete, submittable financial-fraud complaint in under one minute. Words typed: zero. English required: none. Figure 2 overleaf shows the same journey as a single view.</i>		

7. What Makes It Different

- **It is not a chatbot.** Four fixed questions, recognition-only inputs, no free-text conversation anywhere in the flow.
- **It is not a translation layer.** A translated bad form is still a bad form. Rok deletes the questions instead.
- **It is not “add voice”.** Voice has already been added to this exact form by Karnataka Police, precisely because the form is too slow. Rok’s accessibility comes from the input modality being recognition of the user’s own evidence — which is simultaneously the audio path, the text path, the icon path and the fast path.
- **It has a different unit of success.** An opened case plus one freeze-relevant identifier, rather than a completed form.
- **It designs for the helper.** Assisted reporting is treated as the normal case, not an edge case.
- **It designs for fear.** The “still on the call” branch and the never-asks-for-an-OTP guarantee treat the victim’s psychological state as a first-class input, not as a tone-of-voice problem.
- **It cannot fail halfway.** The case exists from the first tap. Abandonment produces a lead, not a void.

Figure 3 overleaf sets the two experiences side by side.

FIGURE 2

The 60-Second Journey

Kamala, 68, Chennai. Tamil speaker. Rs 45,000 taken at 11:47 PM. Words typed: zero. English required: none.

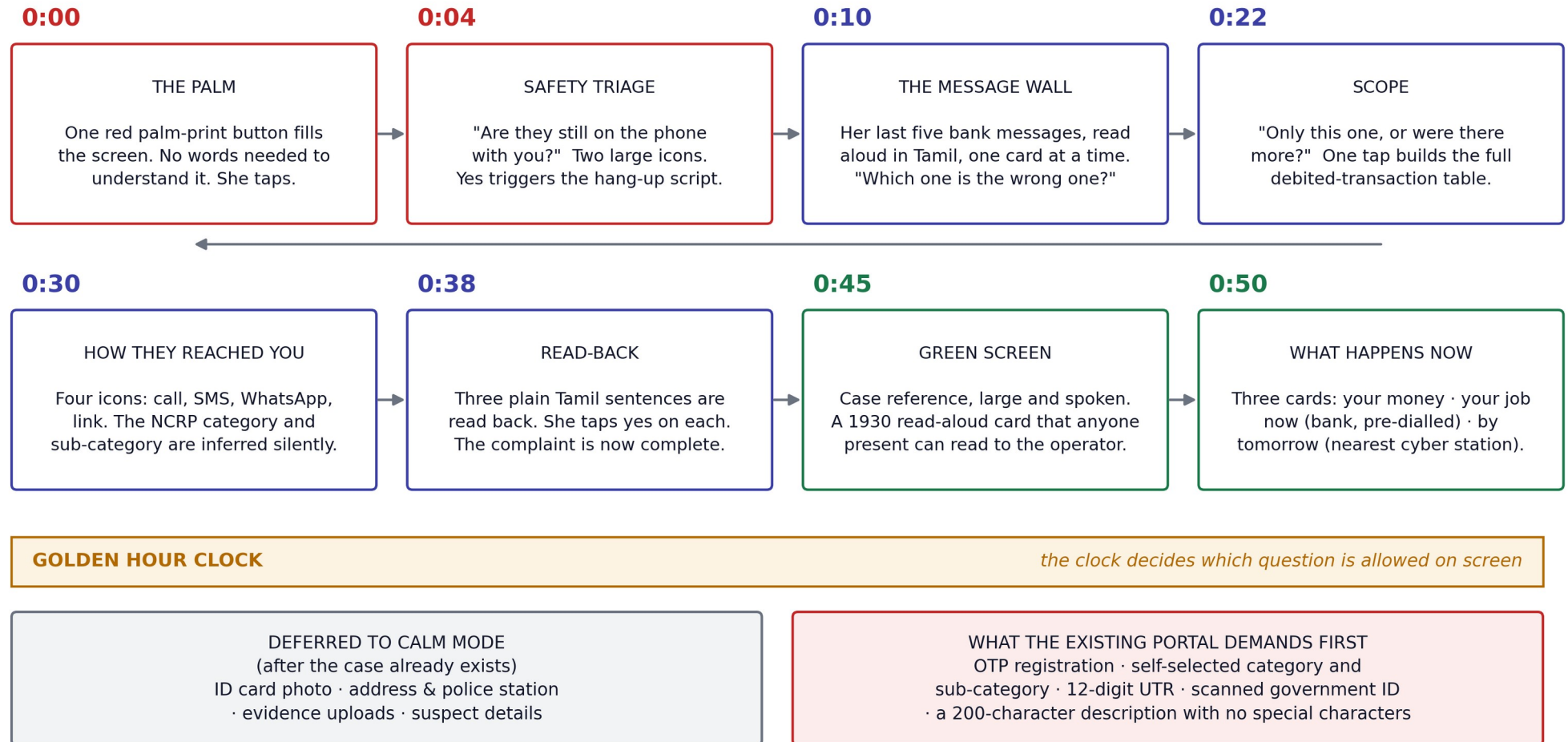


Figure 2 — The 60-second journey. The Golden Hour clock permits only the questions that change whether the money is frozen; everything else is deferred to calm mode.

FIGURE 3

The Inversion

NCRP is a prove-it document, structured for the investigator who reads it later. Rok is a stop-it signal, structured for the money that is moving now.

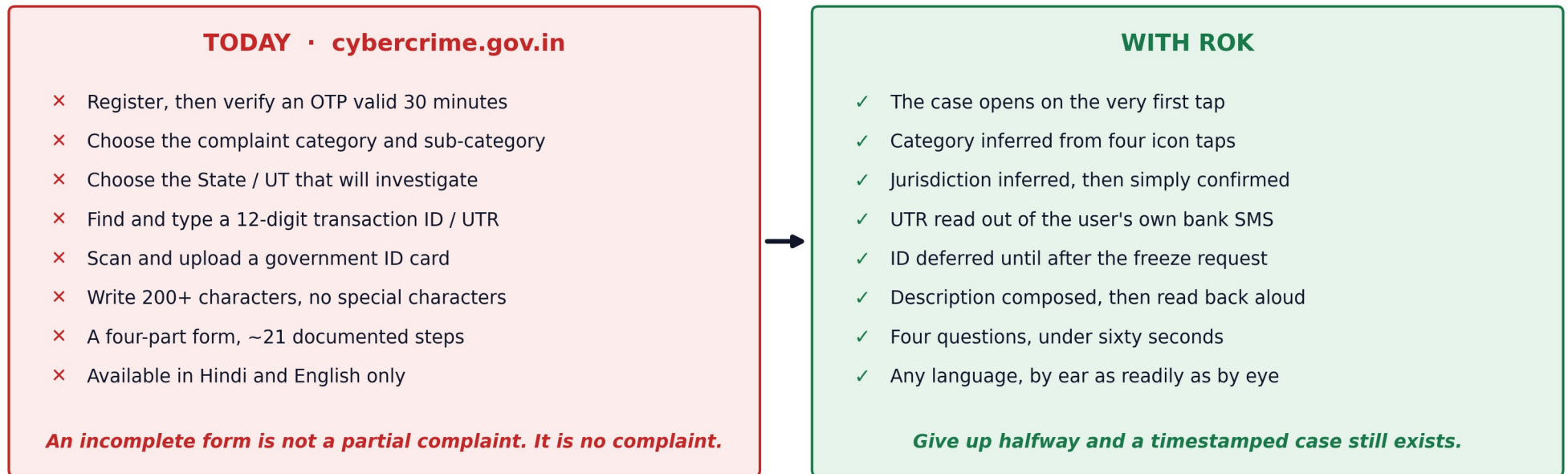


Figure 3 — The inversion. The same nine minutes of a victim's life, structured for the investigator on the left and for the money on the right.

8. Why This Could Stand Out in the Hackathon

- **The stakes are legible in ten seconds.** No judge needs a slide to understand “₹45,000 and a clock”.
- **The demo is a race.** Split screen: the real cybercrime.gov.in flow on the left, Rok on the right, one timer across both. This is the most watchable demonstration available across any of the ten platforms.
- **It is the only brief where accessibility and speed are the same requirement.** In most redesigns, accessibility is a tax on the mainline experience. Here the accessible path is the fastest path — which is the strongest possible argument for universal design, demonstrated rather than asserted.
- **Every claim is government-sourced.** The 200-character rule, the UTR requirement, the two-language limit, the NVDA link and the ₹4,005 crore lost by senior citizens all come from cybercrime.gov.in itself or from replies placed before Parliament.
- **It is adoptable tomorrow.** It does not ask I4C to rebuild NCRP. A bank branch, a CSC operator or a state cyber cell could run it as an intake front end without any change to the portal.
- **It is honest about its boundary.** The POC does not fake an API it does not have, and says so on screen. Judges reward that and punish the opposite.

9. Browser POC Scope

Fully client-side, single-page, no backend — which is also the privacy story: nothing leaves the browser until the user presses submit.

Built for real	Detail
Recognition-first flow	Complete state machine, all six mechanics, live Golden Hour clock driving question gating.
Evidence Recognition Engine	Deterministic parser for Indian bank debit-SMS formats (SBI, HDFC, ICICI, Bank of Baroda, PNB, and UPI app alerts) → amount, timestamp, bank, account tail, UTR/RRN, beneficiary VPA, each with a confidence flag.
Screenshot path	Image upload → OCR → the same parser. This is the genuinely working real-device path.
Speech	Web Speech API: speechSynthesis for output and SpeechRecognition for the optional spoken trigger, in ta-IN, hi-IN and en-IN.
Narrative composer	Auto-generated ≥200-character description with special characters stripped, plus sentence-by-sentence read-back.
Taxonomy inference	Four icon taps mapped to the real NCRP category and sub-category values.
Accessibility	Full keyboard path, ARIA-live regions, visible focus, 200% zoom tolerance, verified end-to-end with NVDA.
Outputs	1930 read-aloud card, NCRP-shaped JSON payload, printable case PDF, guardian handoff link/QR.

Simulated — and labelled as such on screen	Why, and what the real path would be
The phone SMS inbox	Browsers cannot read SMS. The demo loads a fixture inbox from JSON. On a real device this is an SMS-read permission or a forward-to-shortcode. The screenshot upload path is fully real and is demonstrated live.

Simulated — and labelled as such on screen	Why, and what the real path would be
Submission to NCRP	There is no public NCRP API. The POC produces a validated, ready-to-submit packet — not a fake success screen. State this out loud during the demo.

Stack	Choice
Front end	React + Vite, single page, no router. Tailwind for layout.
State	A single explicit finite state machine — the flow must be auditable, not emergent.
Parsing	Hand-written deterministic regex ruleset per bank template. No model in the critical path.
OCR	Tesseract.js, in-browser.
Speech	Web Speech API, with pre-recorded audio fallback for the ~30 fixed strings in the demo languages.
PDF	Client-side generation; no server round trip.
Persistence	Browser-local only. No credentials, no account numbers, no transmission.

10. Key Screens and Interactions to Build

Ordered by build priority. Screen 3 is the one that decides whether the idea lands — build it first and build it beautifully.

#	Screen	What to build	Priority
1	The Palm	One full-bleed button, comprehensible without reading. Language auto-detected. Microphone permission requested only on tap.	P0
2	Safety Triage	“Are they still on the phone?” Two icons, spoken. Full hang-up and alert-a-relative branch.	P0
3	The Message Wall	The signature screen. Large SMS cards, one-at-a-time audio, tap-to-select, extracted fields animating into a receipt.	P0
4	Scope	“Only this one, or more?” Multi-transaction handling.	P0
5	How They Reached You	Four icons, silent taxonomy inference. Show the inferred NCRP category in a debug ribbon for the judges.	P0
6	Read-Back	Three sentences in the user’s language, yes/no on each, audio and text simultaneously.	P0
7	Green Screen	Case reference, very large. The 1930 read-aloud card. The three next-step cards.	P0
8	Guardian Handoff	QR / short code. Demonstrate with two browser windows side by side.	P1
9	Calm Mode	Deferred ID upload and evidence attachment — deliberately slow and gentle, to make the contrast visible.	P1
10	The Race	Side-by-side comparison view built for the pitch itself.	P2

11. Risks and Limitations

Risk	Sev.	Mitigation
No public NCRP API. A POC cannot actually file the complaint.	High	Do not fake it. Ship a validated packet plus a 1930 script, and frame Rok as an intake layer that I4C, a state cyber cell, a bank branch or a CSC operator could adopt as-is.
SMS parsing is brittle; Indian bank formats vary and change.	High	Never auto-submit. Always read back. Visually flag low-confidence fields. Always allow manual correction.
A wrong identifier could cause an innocent account to be frozen.	High	Explicit human confirmation before submission. Never infer a beneficiary the user has not confirmed. Log field-level provenance.
Rok's own design could be cloned as a second-stage recovery scam.	High	Never request an OTP, PIN, password or full account number. State this permanently on screen. In production, distribute only through official channels.
Web Speech API language coverage is uneven across browsers.	Med	Pre-recorded audio for the ~30 fixed strings in the demo languages. Treat TTS as an enhancement, never a dependency.
Recognition assumes clean evidence exists.	Med	Card-present fraud, cash handovers and multi-day digital-arrest cases have no single clean debit SMS. Rok degrades to a guided manual path — still faster, but with less of the magic. Say so rather than claiming universality.
It cannot fix what happens after intake.	Med	Investigation quality, recovery rates and the FIR process are out of scope. Rok buys minutes, not justice. Claim only the minutes.
Trust: a fraud victim is being asked to trust another website.	Med	Nothing worth phishing is collected. Client-side only. A visible, permanent statement of what is never asked for.

12. One-Sentence Pitch

THE PITCH

Rok turns India's cyber-fraud complaint from a 21-step form you must complete into a single question you can answer — "which of these messages is the wrong one?" — so that a 68-year-old who reads no English can get a freeze request moving in under sixty seconds, using nothing but her own bank SMS and her thumb.

Appendix A — Evidence Register

Every claim in this document that describes the current system is sourced. No statistic here is estimated, extrapolated or invented.

Claim used	Source
Mandatory: incident description of minimum 200 characters without special characters; national ID soft copy; 12-digit transaction ID / UTR; bank / wallet / merchant name; transaction date; fraud amount	cybercrime.gov.in — pre-login instructions page
Registration by name and Indian mobile number with an OTP valid for 30 minutes; category and sub-category selection; complaints handled by State/UT police based on the complainant's own State/UT selection	cybercrime.gov.in — official FAQ
Portal language options are Hindi and English; accessibility affordances are contrast toggles plus a link to download NVDA	cybercrime.gov.in — site chrome, observed directly
Four-part complaint form (Incident Details, Suspect Details, Complaint Details, Preview & Submit), documented as roughly 21 steps	Citizen procedure manuals published by Bank of India and L&T Finance
1,03,488 senior citizens and 4,63,114 women lodged financial cyber fraud complaints totalling ₹7,769.63 crore in 2025; senior citizens' share ₹4,005.12 crore	Ministry of Home Affairs, written reply, Rajya Sabha (August 2026)
More than ₹7,000 crore saved across more than 23 lakh complaints through CFCFRMS; CFCFRMS launched 2021 for immediate reporting and to stop siphoning of funds	Minister of State for Home, written reply, Rajya Sabha (December 2025)
Cyber-fraud losses rose from ₹7,465 crore (2023) to ₹22,845 crore (2024); cases from 24 lakh to 36 lakh	MHA data placed in Parliament, July 2025
1930 helpline upgraded with IVR and a voice-guided WebBOT; if a caller cannot reach staff or disconnects due to long waits, an SMS link to the WebBOT is sent and staff later enter details into NCRP manually	Karnataka State Police announcement, April 2025
Digital-arrest fraud method: isolation of the victim on a continuous call, psychological coercion, instructions not to inform anyone, secrecy oaths	Documented case reporting, Karnataka and Chandigarh, 2025
Elderly couple in Karnataka died by suicide in March 2025 after being defrauded of ₹50 lakh	Reported case, March 2025
Elderly share of reported cybercrime cases in Goa: 6.17% (2023) → 14.11% (2024) → 12.21% (2025)	Goa Police data, reported January 2026

Sources were retrieved in August 2026. Figures reported in Parliament are as stated by the Ministry; they are reproduced here without adjustment.

Appendix B — Field Mapping

What NCRP demands, against how Rok obtains it. The right-hand column is the whole argument.

NCRP mandatory field	How the citizen supplies it today	How Rok obtains it
Complainant name and mobile	Typed, then OTP-verified	Typed once, or carried by the guardian handoff. Deferred if the case is opened anonymously.

NCRP mandatory field	How the citizen supplies it today	How Rok obtains it
Category of complaint	Selected from a dropdown	Inferred from one icon tap
Sub-category of complaint	Selected from a dropdown	Inferred from the same icon tap plus the parsed transaction type
State / UT	Selected by the complainant	Inferred from device locale, then confirmed by a single yes
Incident date and time	Typed	Read from the timestamp inside the selected bank SMS
Fraud amount	Typed	Parsed from the selected bank SMS
Bank / wallet / merchant	Typed or selected	Parsed from the sender ID and body of the selected bank SMS
12-digit transaction ID / UTR	Located and typed	Parsed from the selected bank SMS or from the uploaded screenshot
Victim account / card / wallet ID	Typed	Account tail parsed from the SMS; the full number is never requested
Incident description (≥200 chars, no special characters)	Written by the victim	Composed by Rok from confirmed facts, then read back sentence by sentence for a yes or no
National ID soft copy	Scanned and uploaded	Deferred to calm mode, after the freeze request is out
Supporting evidence	Uploaded	Deferred to calm mode; the selected SMS is retained automatically as the first exhibit

Appendix C — Build Plan

Block	Focus	Definition of done
Hours 0–4	Parser and fixtures	Deterministic SMS parser passing against a fixture set of at least 25 real-format Indian bank debit alerts, with confidence flags.
Hours 4–10	The Message Wall	Screen 3 complete: card list, per-card audio, tap-to-select, extraction receipt animation. Keyboard and NVDA verified.
Hours 10–16	The spine	Screens 1, 2, 4, 5 wired into the state machine. Clock gating live.
Hours 16–22	Composer and read-back	Screen 6 complete. ≥200-character output validated against NCRP's character rules.
Hours 22–28	Outputs	Green screen, 1930 card, NCRP packet, PDF. Screenshot/OCR path working end to end.
Hours 28–32	Guardian handoff and calm mode	Screens 8 and 9. Two-window demo rehearsed.
Hours 32–36	The Race and the pitch	Side-by-side comparison view, accessibility pass with NVDA and keyboard only, demo script timed under three minutes.

Rok · रोक · *Stop the money first. The paperwork can wait.*